

11

Monoclonal Antibody Continuous Processing Enabled by Single Use

Mark Brower, Ying Hou, and David Pollard

11.1

Introduction

The growth of therapeutic proteins and vaccines continues to gain momentum in the marketplace with biologically derived molecules owning a predominant position among the top 10 selling drugs. Biotherapeutics remains the largest and fastest growing class of novel pharmaceuticals entering the market where in 2012 approximately 28 biomolecules were approved and 350 biomolecules were in the clinical development phase [1]. The annual growth is expected to continue at a rate of 9.5% for the period between 2010 and 2016 [2] equating to a US \$58 billion industry [3]. While the economic success has been dominated by full-length humanized monoclonal antibodies (mAbs), expansion into new modalities is accelerating including antibody drug conjugates, multispecific antibodies, glycan engineered proteins, and novel fragments and scaffolds.

Despite this growing success, the biopharmaceutical industry faces strict competitive challenges from multiple pressures including economic, regulatory, and political [4]. As a result, there is a significant drive to boost the overall productivity of biotherapeutic programs by shortening development timelines and lowering both development and production costs, while maintaining product quality [4]. Recently, progress has been made to significantly accelerate process development timelines by innovative, high throughput, technology solutions for the cell line development and upstream process (USP) characterization spaces [5,6]. High throughput methods for chromatography development have also been demonstrated in the downstream space [7]. However, the inherent inefficiency of those batch-based chromatographic separations leads to relatively unproductive downstream purification processes (DSP) as a whole.

The expanding diversity of biotherapeutic pipelines with novel modalities is pressuring the industry away from the traditional fixed 10–20 kl stainless steel manufacturing facilities built in the last two decades toward more flexible approaches that can easily accommodate both large and small drug demands from the diverse modality platforms each of which may require specialized technologies in the production process. This evolving competitive business

environment is driving the biopharmaceutical industry toward a tipping point for radically improved, lower cost bioprocessing options [8]. Because of the large number of full length antibodies in development, and their relatively high dosing requirements, low-cost mAb production solutions must be aggressively pursued. Therefore, the strategies presented herein are focused on monoclonal antibody production; however, the basic principles can be applied to other therapeutic protein classifications.

11.1.1

Single-Use Revolution to Enable Process Intensification and Continuous Processing

The benefits of single-use technology with disposable components is well understood to provide the much needed flexibility while significantly reducing the up-front investment in a facility [9]. The elimination of clean- and sterilize-in-place procedures reduces equipment complexity that can dramatically lower capital cost, facility footprint, and increase turnover time/throughput [10]. For example, the economic analysis presented in Figure 11.1 shows that up to a modest

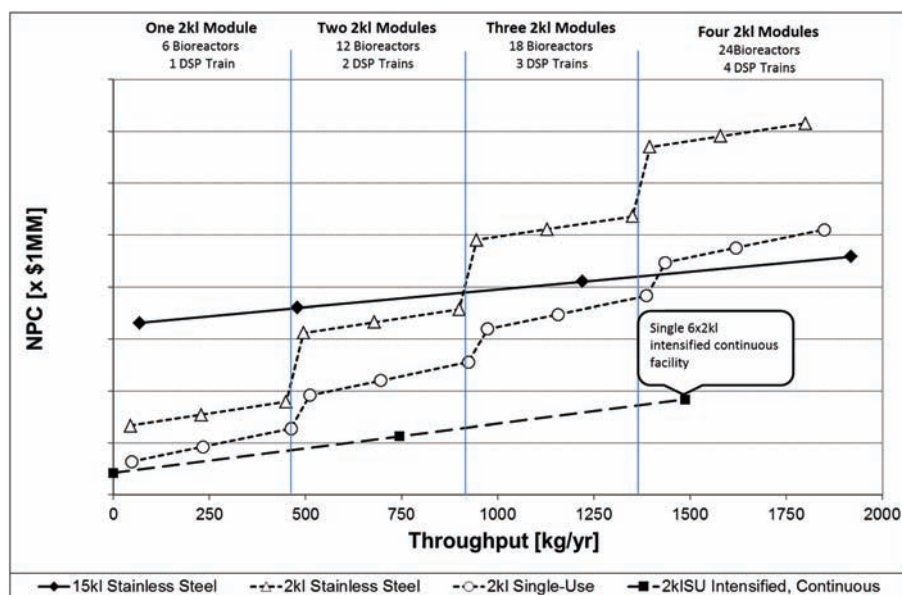


Figure 11.1 Comparison of single-use (complete end-to-end process) and stainless steel facility economics for the production of monoclonal antibody drug substance assuming a fed-batch bioreactor titer of 3 g l^{-1} . It is noted that the lowest value of net present cost (NPC) is the preferred outcome and recognizes the total expense associated with the facility

including construction, capital, and operating costs for a brownfield facility assuming 20 years of operation. A 15 000 l stainless facility base case is compared to: one module consisting of six bioreactors and one purification train at 2000 l scale either stainless or single use. The capacity is increased by adding identical 2000 l modules.

1300 kg year⁻¹ and at 3 g l⁻¹ upstream titer the fully single-use mAb facility at 2000 l is the preferred option (i.e., lowest net present cost) over either the 2000 or 15 000 l stainless steel configurations. At higher upstream titers, such as 10 g l⁻¹, the 2 kl single-use is always the lower-cost option compared to stainless steel. Furthermore, intensified continuous processes represent a game-changing strategy, which shows economic advantages on the order of net present cost (NPC) difference NPC of \$200MM over the lifetime of a facility.

This realization has driven the industry to define upstream intensification processes [11] to concentrate product throughout the course of the cell culture. Here, an ultrafiltration membrane is used on the bioreactor to retain cells and product while media is perfused through the bioreactor. This has taken the molecular biology and fermentation advancements in fed-batch expression from 3–5 g l⁻¹ titers to intensified processes with titers of 10–15 g l⁻¹. As such a 2000 l single-use bioreactor with intensification can generate a similar volumetric productivity as a 6000 to 10 000 l stainless steel bioreactor. This substantial increase in upstream productivity must be matched by the downstream to realize highly productive protein production facilities. To date however, protein purification processes have been comprised of relatively low productivity unit operations, such as batch-based chromatography (~15 g l⁻¹ h⁻¹) and tangential-flow filtration strung together in a linear manner resulting in relatively low throughputs. As bioreactors

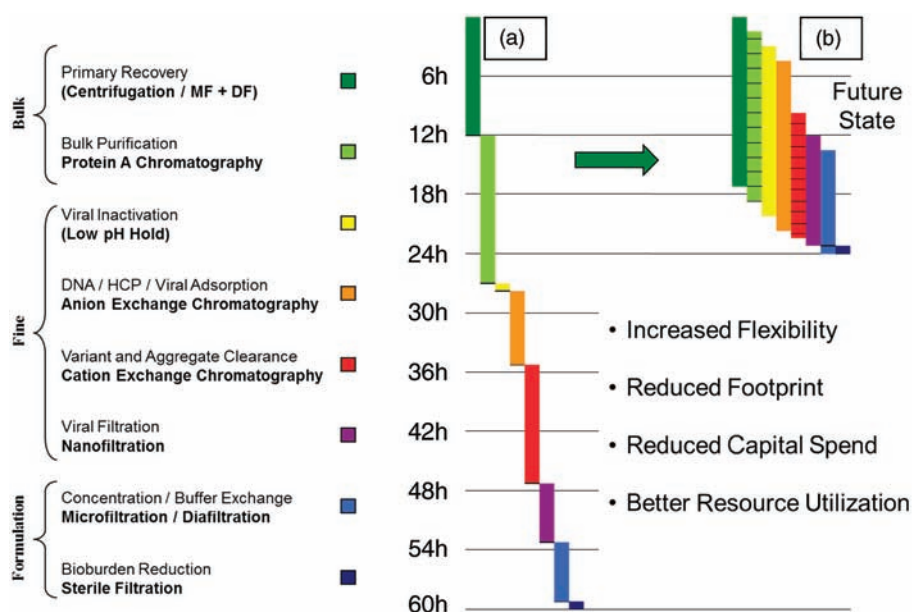


Figure 11.2 Example of (a) batch-based DSP processing time with linear processing for a generic three-column purification process as (b) the productivity gain, which can be realized if all of the unit operations are run concurrently.

have become smaller with intensified processing, the chromatography columns have become larger with multiple cycles and longer processing times that impact the facility output [12]. Therefore, obvious benefits exist for an intensified, continuous process that can handle changing upstream capacity without dramatic changes in chromatography column size and processing time.

In a standard three-column monoclonal antibody purification process shown in Figure 11.2a) it is common practice to pool the entire product stream from the preceding unit operation prior to moving on to the next. This not only requires the installation of large interoperation holding vessels where the intermediate is sampled for purity and quality, but also leads to lengthy processing times, on the order of 40–80 h in-suite, before arriving at purified drug substance. A future state is also presented in Figure 11.2b with a fully continuous downstream purification train. This future state offers increased flexibility, reduced footprint, reduced capital expenditure, and better resource utilization compared to traditional batch processing in stainless steel facilities.

A high-level gap analysis of traditional batch-based downstream processing is presented in Figure 11.3 where nondisposability and low productivity are

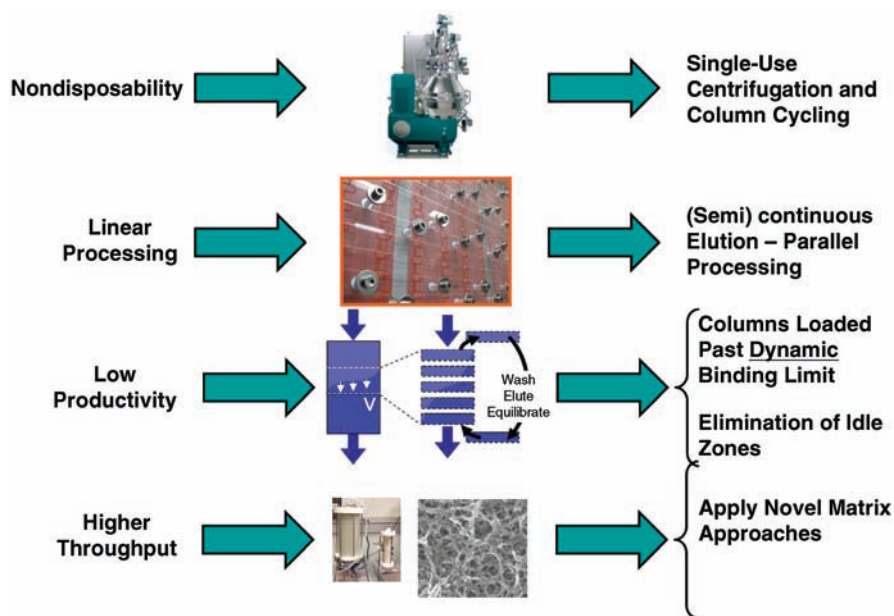


Figure 11.3 A summary of the existing challenges to batch processing and answers that provide a continuous solution: (1) Single-use centrifugation allows a disposable flow path for cell removal; (2) CMCC incorporates heavy column cycling, which transforms columns into single-use components and allows for

an integral approach to the product pool; (3) countercurrent column loading of CMCC increases the specific productivity of chromatography steps; and (4) next-generation single-use matrices that allow for further specific productivity enhancements in chromatography steps.

identified as key technological challenges. With the use of emerging technologies designed to fill these gaps, such as continuous single-use centrifugation integrated with continuous multicolumn chromatography (CMCC) and single-pass tangential-flow filtration (SPTFF), the fully continuous DSP becomes a feasible option. This enables the continuous processing benefits to be recognized, such as short processing times, reduced capital cost, high productivity, and streamlined process [13]. One chief obstacle in transitioning from batch to continuous operations is the cleaning and sanitization validation on complex dedicated continuous processing skids, which may require such vigor as the complete disassembly, cleaning, and reassembly of hardware. With the use of fully single-use wetted paths, including the highly cycled CMCC chromatography columns, this concern is largely alleviated. Additional productivity benefits are realized from the elimination of lengthy clean-in-place and sanitize-in-place steps, as well as from loading enhancements made during the CMCC steps (see Section 11.1.2).

The evolution of the DSP to continuous processing can be thought of as stepwise transformation from (i) the current 15 kl scale batch-based processes to (ii) 2 kl intensified, single-use enabled, to (iii) continuous end to end processing, to finally (iv) commodity processing with next-generation materials and streamlined DSP train as seen in Figure 11.4. The envisioned future state with a fully single-use, continuous purification approach provides a flexible low-cost solution to integrating all the unit operations in a closed process. Closing the system allows facility air handling classification to be reduced compared to current practices (Grade C and B space) as well as transitioning away from individual containment suites to a low-cost, single-room suite

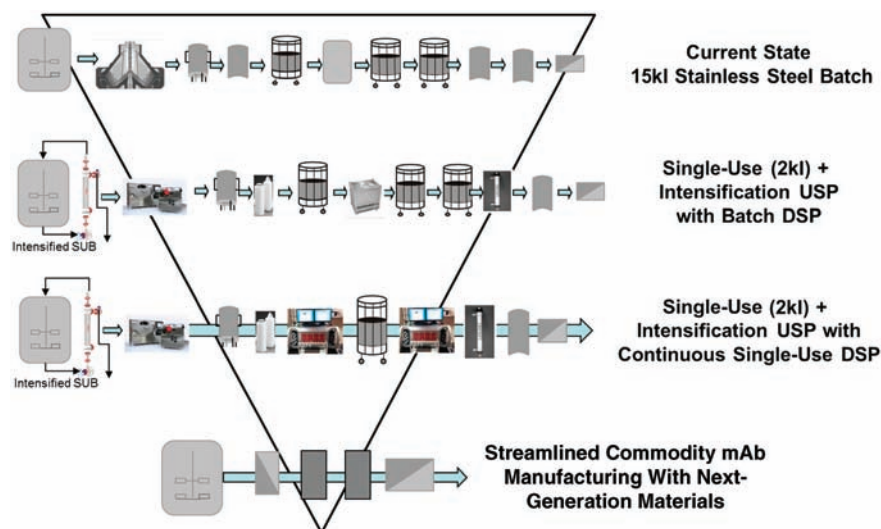


Figure 11.4 The transition from batch processing to continuous enabled by single-use technologies.

housing multiple unit operations, which ultimately facilitates a multiproduct facility [14].

11.1.2

Principles of Continuous Multicolumn Chromatography for Biological Production (BioSMB)

The proposed continuous downstream process relies most heavily on transforming batch-based chromatographic steps into a pseudo steady-state process with periodic column elutions. As such, this section provides a summary of the principles governing continuous multicolumn chromatography operation and the chromatography skid used to perform the methods described herein (Tarpon Biosystem BioSMBTM). A more detailed description can be found in Chapter 10 of this book and by Bisschops. For clarification, CMCC steps are referred to as continuous due to the uninterrupted flow of feed, not to imply that a continuous stream of product is generated.

Classic steady-state countercurrent chromatography, a subset CMCC process and termed simulated moving bed (SMB) chromatography, has been adopted for industrial applications including amino acid, sugar, and chiral separations to overcome capacity bottlenecks [15–18]. In these systems, a stationary phase flow in a direction opposite to the isocratic mobile phase flow is simulated through valve switching between columns connected in a circuit. The result is a steady stream of enriched extract and depleted raffinate leaving different columns in the circuit due to differences in the partition coefficient for the stationary phase for each component in the binary feed solution.

Typically the adsorbent flow is simulated by switching column inlet and outlets by sophisticated stainless steel valve manifolds up to 100 000 l in scale in industries where sanitary processing is not a requirement. These applications have shown 20% resin reduction and 50–55% buffer savings [19]. The benefits of reduced resin and buffer usage can now be applied to bind and elute biologics systems by implementation of a single-use wetted path including a zero-dead volume valve manifold system from Tarpon Biosystems Inc. [20]. The valve block accommodates 240 individually addressable diaphragm valves that provide asynchronous switching and other flexibility in method generation. The single-use aspect of the wetted path negates the need to validate cleaning of the pump heads, valve block, or sensor array between batches or campaigns.

In contrast to a classic SMB process, bind and elute chromatography common to the biopharmaceutical industry is not typically operated isocratically (as is the case with affinity and ion-exchange chromatography), and as a result, it is necessary that a column must undergo discrete steps within the chromatography process, such as loading, washing eluting regeneration, and equilibration. To maximize the dynamic binding capacity and specific productivity of bind and elute steps, the user creates countercurrent flow in only the “loading zone” where simulated stationary flow is in the opposite direction to the incoming crude feed stream using a comparatively small columns number (ranging from

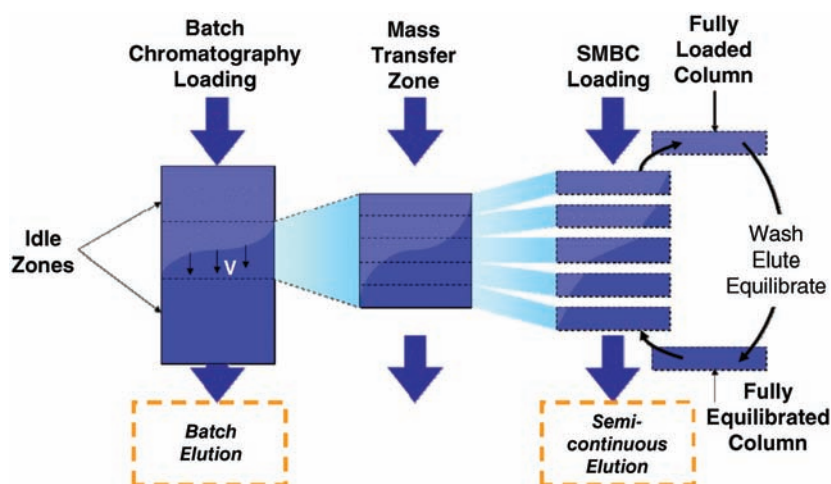


Figure 11.5 Comparison of the loading zone from batch chromatography where idle mass transport zones are present to continuous multicolumn chromatography where those

idle zones are eliminated by allowing product breakthrough in the leading column(s) to be captured by the next small column in series.

2 to 6). As such, the first column in a series in this zone is loaded to saturation before the feed stream is switched to the next column in series to continue loading while the first column undergoes the washing and elution cycle. The other steps in the chromatographic process are then carried out on an individual column, or columns, through asynchronous valve switching. Upon regeneration and equilibration, the column is finally reintroduced to the loading zone as the last column in the series [21].

During the loading step in traditional batch chromatography, a mass transfer zone develops as a wave front traveling through the column, which corresponds to a small fraction of the entire column volume as shown in Figure 11.5. In front of the mass transfer zone, the chromatography media remains saturated and at equilibrium with the feed solution so limited mass transfer occurs. Beyond the mass transfer zone, the feed solution is lean in product and again, the mass transfer is limited. As such a majority of the column is underutilized for the physical adsorption process during the time spent loading the column, which contributes to the low specific productivity of chromatographic separations. In addition, the column loading is typically stopped well before the mass transfer zone reaches the outlet of the column to prevent product loss to the flow-through fraction. This leaves a significant portion of the static binding capacity of the column underutilized such that large-scale columns may only be loaded to 65% of their potential capacity, another contributing factor to low specific productivity [20]. In contrast, countercurrent contact of the feed solution with the stationary phase in CMCC applications provides improved driving force for mass transfer, which is now occurring over the entire bed volume of the loading

zone [20]. Practically speaking, this requires that the first column(s) in the loading zone is loaded past its dynamic binding capacity and as a result, product will be found in the effluent. This product, which would have been lost to the flow-through fraction in a batch process, adsorbed in the next column in series in the loading zone of CMCC. As such, specific productivities of for CMCC range from 2 to 10× compared to an equivalent batch-based process.

The BioSMB skid not only facilitates the countercurrent flow in the loading zone, but also carries out other chromatographic steps on separate columns simultaneously. These complex methods can be visualized using a circular chronogram as shown in Figure 11.6, where the outer ring displays the feed to each individual column and the inner ring shows where the effluent of each column is directed. In the example presented in Figure 11.6 for an eight-column affinity adsorption example, the loading zone is shown in green where crude feed is introduced to three columns starting with column 6. The effluent of column 6 is fed to column 7, the effluent of column 7 is fed to column 8, and the effluent of column 8 is sent to the flow-through fraction. All other chromatographic steps in the process are occurring simultaneously.

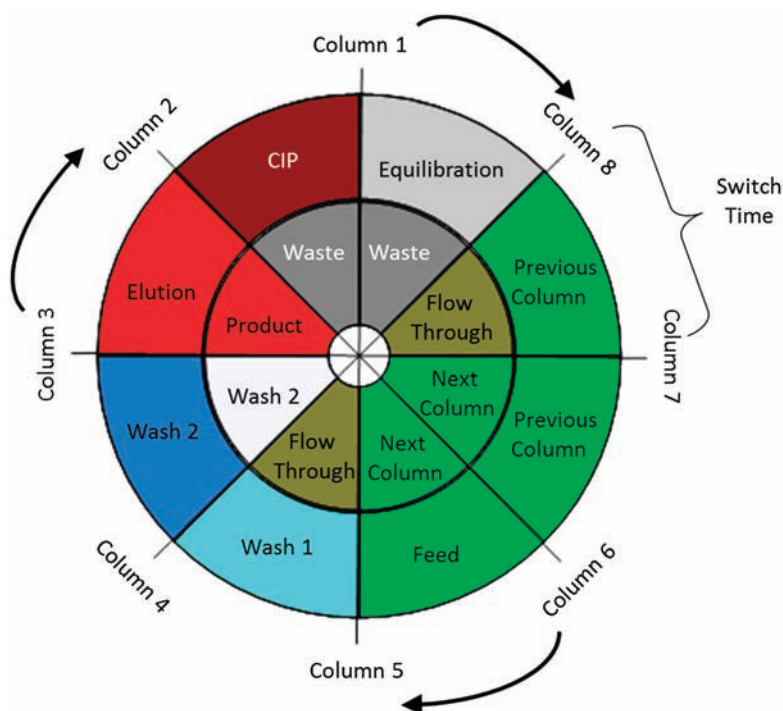


Figure 11.6 Circular chronogram example for zone (green) and columns 4 and 5 are in the bind and elute step with eight columns. The wash zone and so on. The inner ring shows the effluent of a column is directed. The outer ring shows the feed solution of each column with columns 6, 7, and 8 in the loading

The time it takes for a column to switch positions in the load zone is referred to as the switch time (t_s) and can be calculated from Equation 11.1.

$$t_s = \frac{(DBC_{CMCC} \times V_{col})}{Q_F \times c_0} \quad (11.1)$$

where DBC_{CMCC} is the dynamic binding capacity achieved in the CMCC loading zone, V_{col} is an individual column volume, Q_F is the feed flow rate, and c_0 is the concentration of product in the feed.

A full process cycle requires that all columns spent must undergo all of the steps in the chromatographic process and is equal to the number of columns in a CMCC process times the switch time. The time dedicated for each nonloading step is based upon the required buffer volume and a flow rate limitation. An additional constraint imposed on the system is that all nonloading steps must be completed in a time referred to as residual processing time (RPT), which can be calculated from Equation 11.2. This constraint may require that the loading steps are run at different linear velocities from the remaining steps of the process.

$$RPT = (Col_T - Col_L) \times ST \quad (11.2)$$

where Col_T and Col_L refer to the total number of columns in the process and the number of columns in the loading zone, respectively.

Incremental changes in the upstream titer can be compensated for in the CMCC process by adjusting the switch time as per Equation 11.1. If large changes in the titer occur additional columns of the same size can be added to the process as per Equation 11.2 to keep the RPT consistent resulting in similar process kinetics between a 2 g l^{-1} or 10 g l^{-1} feed stream. Typically a design model is created for the capture step that relates capture efficiency to the required resin volume and flow rates in the load zone that allows optimization of specific productivity. Typical models show that 2–4 columns can be used in the load zone with using 6–8 total columns for a 3 g l^{-1} titer process or between 8 and 10 columns for a 10 g l^{-1} process. The CMCC system offers a degree of freedom in the processing time (i.e., Q_F) to accomplish the same separation. The user may choose to minimize the column volume (and thus operate at a large number of cycles for a given separation), or to minimize processing time (and thus increasing column size).

11.2

Continuous Downstream Processing for Monoclonal Antibodies Unit Operation Development

To realize a truly continuous, integrated downstream purification process each unit operation feeds directly into the next as summarized in a block-flow diagram shown in Figure 11.7. All of the familiar unit operations are represented starting with crude cell culture broth from a single-use bioreactor,

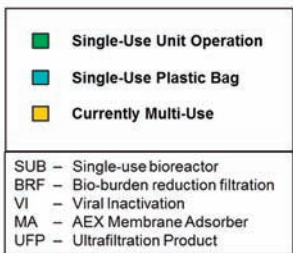


Figure 11.7 Block flow diagram of a continuous downstream purification process. All of the unit operations in the downstream are run at the same time using surge vessels as a place for stream conditioning for feed to the next step.

which is fed through a single-use centrifuge to remove biomass, and then a bank of depth and sterilizing grade filters to remove remaining particulates from the centrate. The clarified feed stream is then processed by a CMCC affinity protein A capture step for bulk purification. The protein A effluent is treated to a low pH viral inactivation step for a specified period of time prior to loading on an anion exchange membrane. The anion exchange product is polished through a second capture chromatography step prior to viral filtration. The viral filtrate is concentrated to an intermediate concentration using single-pass tangential-flow filtration and the retentate is collected for a specified period of time. Finally, the ultrafiltration product (UFP) is diafiltered into formulation buffer in discrete lots, which can be used as a point for batch definition. Due to the interdependency of each unit operation on the others, the individual steps were studied and characterized around a center point separately (described in Sections 11.2.3–11.2.6) before integrating them together for the demonstration of a continuous process.

This continuous process is said to be at pseudo steady state after a startup period required for feed to reach each individual unit operation and before a shutdown period when there is no longer feed available for each unit operation. This pseudo steady state refers only to the flows into and out/of each unit operation and is not intended to assume that the properties of these streams are constant (such as concentration, pH, and conductivity). In fact

gradients in these properties are expected in the effluent of several unit operations. These gradients in the incoming surge stream are largely dampened out by choosing an appropriate volume of surge material to act as a buffer going into the next unit operation. More discussion on this topic will be given in Section 11.2.3.

11.2.1

Surge Vessels and Balancing Flows

In the continuous paradigm the familiar large pooling vessels used in batch operation between each unit operation have been replaced by small single-use mixing bags used as surge vessels. During the pseudo steady state operation, each surge bag must maintain a constant volume where the feed rates to that vessel directly matches take-off rate over a long period of time. Conductivity and pH adjustments for the next unit operation are also made in the surge vessels by adding an appropriate acid, base, or other component for stream conditioning. These additions must be accounted for in take-off rate so that accumulation for each surge vessel is avoided according to Equation 11.3.

$$0 = Q_{\text{out}} - Q_{\text{in}} - \sum_i Q_i \quad (11.3)$$

where Q_{out} is the takeoff flow rate, Q_{in} the incoming flow rate, and Q_i are the flow rates for any component added for stream conditioning.

One factor that aids in the design of the surge vessels for a continuous process is that the target length of each unit operation is increased from 4 to 8 h to the desired continuous processing time. Table 11.1 shows how effluent flows are affected by transforming batch processes into continuous processes.

Table 11.1 Comparison of typical flow rates in downstream unit operations run in both batch and continuous modes assuming a 2000 l bioreactor harvest at 3 g l⁻¹ titer.

Step	Target batch processing time [h]	Batch flow rate [l min ⁻¹]	Continuous processing time [h]	Continuous flow rate [l min ⁻¹]
Centrifugation/depth filtration	4	8	23	1.5
Protein A chromatography	8	15	23	0.5
Anion/cation exchange chromatography	8	8	23	0.6
Viral filtration	8	1	23	0.3
Ultrafiltration (cross-flow rate)	8	25	23	0.3

11.2.2

Primary Recovery: Centrifugation and Depth Filtration

The removal of cells and particulates from cell culture media is traditionally accomplished by stainless steel disc stack centrifugation followed by a relatively small bank of depth and sterilizing-grade filters. For continuous processing limited to the 2000l scale, alternatives to replace the difficult to clean, and capital intensive stainless steel centrifugation are a disposable depth filtration train and single-use centrifugation. A two-stage depth filtration run in series where the pore size of the media is stepped down from stage 1 to stage 2, followed by a sterilizing-grade filter is an effective way to clarify cell culture fluid [22]. This approach to clarification has two limitations. First, depth filter configurations across cell types and specific fermentation processes have proven to be nonplatformable, resulting in a significant rescreening and optimization effort for each new pipeline product. Second, the required membrane area for clarification is proportional to the percentage of biomass in the feed. As a result, intensified fermentation processes with significantly higher biomass will require increasingly more membrane area for an equal volume harvest. Centrifugation provides a more platformable approach and requires minimal optimization for each mAb product.

The single-use kSep centrifuge provides a solution to lengthy steam-in-place cycles utilized in disc-stack centrifuges by providing a fully disposable fluid flow path enabling continuous centrifugation. A typical pilot-scale system consists of 4×100 ml pear-shaped chambers as shown in Figure 11.8 while the scale-up skid for harvesting a 2000l bioreactor consists of 6×1000 ml chambers. The operating g-force of the lab- and pilot-scale kSep centrifuges are $1000 \times g$ and $2000 \times g$, respectively. The centrifuge is prepared in three steps: (1) inserting a gamma-sterilized chamber set into the rotor, which is utilized to collect the cells, (2) inserting a gamma-sterilized tubing manifold into a bank of pinch valves, which directs flows on the skid throughout the automated load and discharge cycles, and (3) aseptically joining the chamber and tubing sets using a tubing welder. The total time it takes a trained operator to turn around the kSep centrifuge is approximately of 0.5 h whereas the cleaning and sanitization cycles for the disc-stack centrifuge is on the order of 8 h.

Cell solids are trapped in the kSep chambers by balancing the centrifugal force, created by the unique carousel design of the rotor, and the drag forces associated by fluid flow into the chamber in the opposite direction resulting in a zero-shear environment. As a result of the balanced forces, cells are retained in a fluidized bed inside each of the chambers, which enables efficient washing of the intracellular space to either the centrate or waste streams once the chambers have been loaded to capacity. Washed cells are flushed out of the chambers to waste by reversing the wash buffer flow into the direction of the centrifugal force. More details on the mode of operation of the kSep centrifuge can be found in Chapter 10 of this book. The kSep centrifugation step is a discontinuous approach where <3 min out of a 10–20 min cycle are spent washing

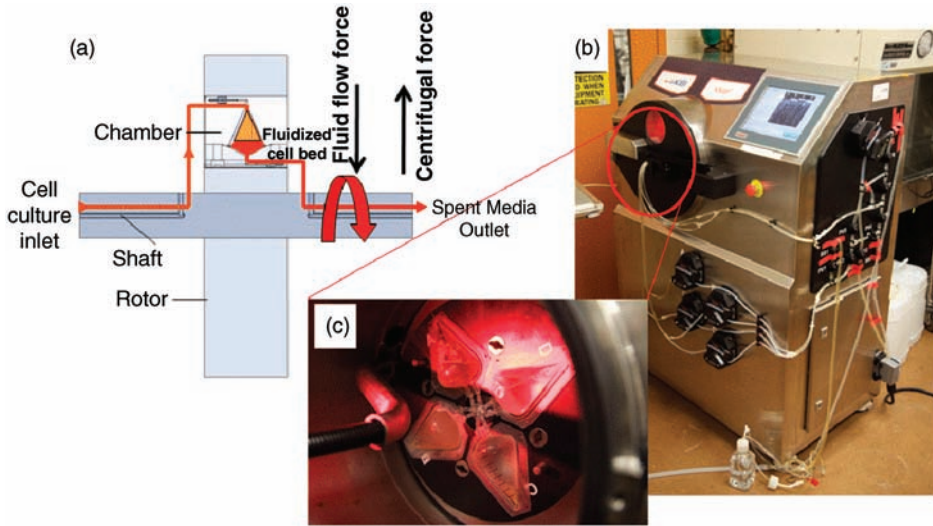


Figure 11.8 Single Use Centrifugation: Panel A describes the mode of cell capture in the kSep centrifuge where the centrifugal force is in the opposite direction to the incoming fluid flow. In this figure, the axis of rotation is in the plane of the page such that the rotor would be

spinning out of the page toward the reader.

Panel B shows the kSep centrifuge in operation with the bank of pinch valves directing fluid flows on the skid. Panel C is a close-up view of the rotor housing the 4 × 100 ml pear-shaped chamber set.

and discharging cells. The time spent when centrate and wash buffer are not being sent forward must be accounted for in flow rate calculation as per Equations 11.4 and 11.5:

$$Q_{\text{kSep Feed}} = \frac{V_{\text{Batch}}}{t_{\text{Batch}} \times \theta_{\text{kSep}}} = \frac{V_{\text{Batch}}}{t_{\text{Batch}}} \times \frac{t_{\text{kSepCT}}}{t_{\text{kSepFeed}}} \quad (11.4)$$

$$Q_{\text{kSep Out}} = \frac{V_{\text{Centrate}} + V_{\text{Wash}}}{t_{\text{kSepCT}}} \quad (11.5)$$

Where Q_{kSep} is the feed-flow rate to the centrifuge; V_{Batch} is the volume of the incoming batch to be processed; t_{Batch} is the desired time to process the batch and θ is the fractional time spent in the loading phase of the centrifugation cycle; T_{kSepCT} is the kSep cycle time and t_{kSepFeed} is the time spent in the feeding phase of the kSep cycle; V_{Centrate} is the volume of centrate collected per cycle; and V_{Wash} is the volume of wash collected per cycle.

Performance of the centrifuge is characterized by the turbidity of the centrate measured in nephelometric turbidity units (NTU). As the chamber reaches its physical limitation for the volume of solids it can hold, particulates start exiting the chamber and causing contamination of the centrate stream. Charting the NTU over loading time allows the operator to determine how many cells a chamber can hold and when to trigger the washing and dumping phases of the

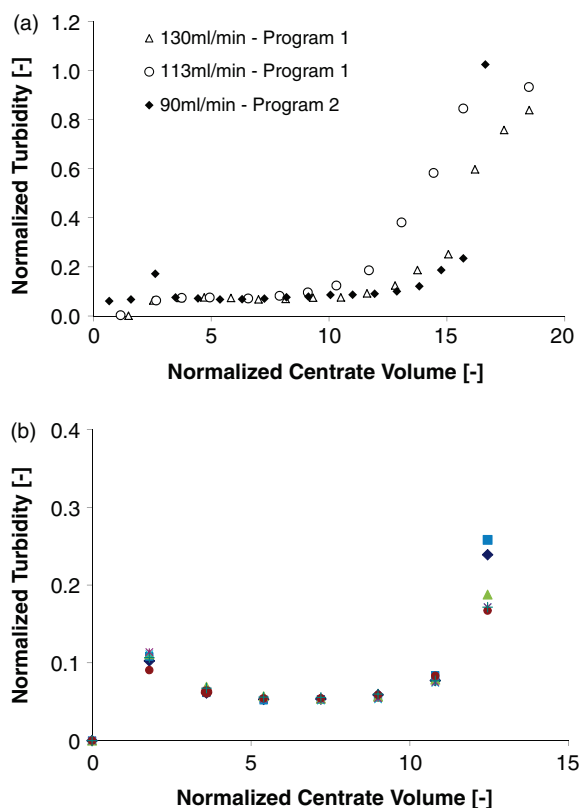


Figure 11.9 (a) Turbidity breakthrough profiles from the kSep centrifuge normalized to the turbidity of the feed stream and to the chamber volume (100 ml) at $1000 \times g$. Two different mAb programs using CHO cells for expression show the similar characteristic breakthrough shape. (b) Seven repeated cycles for program 1 at $90 \text{ ml}^{-1} \text{ min}^{-1}$ chamber⁻¹ flow rate, and $1000 \times g$. Curves in Figure 11.9a and b were generated from different batches of the same program with different initial cell densities.

cycle. These particulate breakthrough curves have a characteristic shape, displaying a period of low turbidity when solids are being efficiently captured, followed by an exponential rise in turbidity once the chamber capacity has been exceeded. Figure 11.9a shows the turbidity breakthrough curves with the 100 ml chamber skid for two different mAb products, each with cell concentrations of $40\text{--}100 \times 10^6 \text{ cells ml}^{-1}$ in the feed. This equates to roughly 10–15 chamber volumes of feed prior to a wash and cell discharge cycle. With the same feed stream (project 1), it can be seen that at a given g-force, higher solids retention and a steeper breakthrough curve is seen at lower flow rates where the centrifugal force becomes more dominant. The separation has been shown to be robust by demonstrating consistent breakthrough curves for >7 cycles as shown in

Table 11.2 Comparison of key operation parameters for the kSep centrifuge and a standard disc-stack centrifuges.

Parameter	kSep performance	Historical disc stack performance
g-force	1000–2000 $\times g$	4000–10 000 $\times g$
Clarification efficiency	>90%	>95%
Centrate pool turbidity	<90NTU	<40NTU
Viability reduction	Negligible over individual cycle <3% over 8 h	Not available
Yield	98%	96%
Turnaround time	0.5 h	8 h
Depth filtration performance	200 l m ⁻² , 220 LMH, <20 psi ^{a)} to	500 l m ⁻² , 150 LMH, <20 psi ^{a)}
Post centrifugation	200 l m ⁻² , 20 LMH, <20 psi ^{b)}	

a) Using a platform depth filter media.

b) Using optimized depth filter media for kSep centrate.

Figure 11.9b and as many as 80 cycles in data not shown. Disc-stack and kSep centrifugation performance is compared in Table 11.2.

The relatively low g-force generated by the kSep centrifuge compared to a disc-stack centrifuge, results in a more turbid centrate pool, which may impact loading volume on the subsequent depth filtration step. Using the same grade depth filter media, the loading was shown to be reduced from of 500 l m⁻² at a constant flux (150 LMH) with disc stack centrate to just 200 l m⁻² using kSep centrate at a similar flux (220 LMH) and the same maximum pressure (20 psi). However, it must be noted that the relatively low g-force of the kSep centrifuge may result in a lower threshold for the diameter of particulates, which are retained by the centrifuge and as a result a more open depth filter media may be more appropriate for use with kSep centrate. Figure 11.10 shows the pressure versus loading profiles seen using the same kSep centrate as feed using two different depth filter media for fluxes of 150 and 20 LMH. These two fluxes were chosen to represent typical first flow rates for batch processing and continuous processing, respectively. It can be seen that the optimized depth filter media outperformed the standard depth filter media at both fluxes. In addition, reduced pressures were observed in the low flux regime consistent with lengthened step times associated with continuous processes. The depth filter effluent is passed through a sterilizing grade filter to remove any remaining particulates in the stream that may plug the downstream chromatography column and to reduce as a means for bio-burden reduction. In conclusion, although the kSep may result in a more turbid centrate stream, there is little impact on the subsequent depth filtration polishing step once an appropriate depth filter medium is selected for the application.

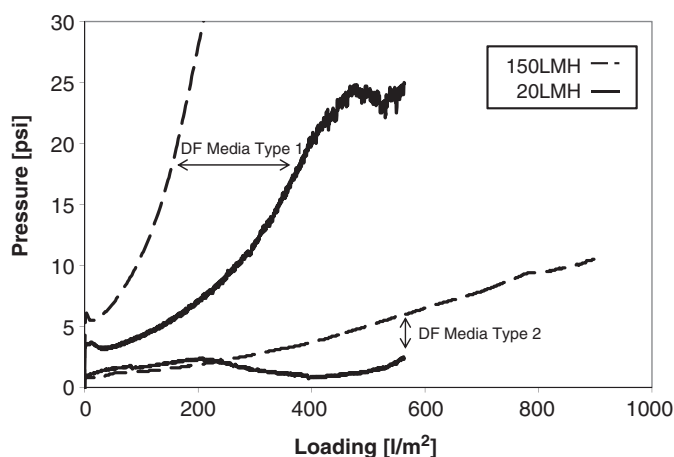


Figure 11.10 Pressure versus loading curves generated using kSep centrate on two different depth filter media. The loading was conducted at constant flow rates at two different flux values: 150 LMH to represent batch type processing, and 20 LMH to represent continuous type processes.

11.2.3

Bulk Purification: Continuous Multicolumn Chromatography – BioSMB Protein A Capture and Viral Inactivation

The clarified cell culture stream exiting the depth filter/sterile filtration train is fed to a Tarpon BioSMB continuous chromatography skid for protein A affinity antibody capture. Protein A is an affinity ligand that captures the constant domain of most monoclonal antibodies subclasses (near neutral pH and over a broad conductivity range). Elution of the product is achieved by an acidic pH shift. Protein A serves as a bulk purification step and raises the antibody purity to >95%.

Due to the increased number of columns in continuous chromatography, and the need to achieve a steady state over a number of cycles, significant feed is required for each experiment compared to traditional batch chromatography carried out on 0.5 or 1 cm diameter packed columns. As such, it becomes critical that the underlying method development be carried out on small batch columns to provide for the necessary separation. Predicated on that notion, continuous chromatography development focuses on transforming an optimized batch method into a continuous method while maintaining the desired separation. Therefore, continuous method development focuses on two main areas – the feeding phase where columns are typically loaded past the capacity demonstrated for batch chromatography, and the elution phase which is critical yield and purity of the product. It may happen that as a result of overloading of the columns, a different purity profile is seen in the CMCC effluent pool after optimization. In these scenarios, the method must be further adjusted to meet the purity targets by altering either how the columns are loaded or

Table 11.3 Batch protein A process used for transformation into a CMCC step.

Step	Buffer	Volume	Flow rate (cm h ⁻¹)
Equilibration	Low conductivity	5CV	300
Loading	Clarified culture fluid	<35 g l ⁻¹	250
Wash 1	Low conductivity	3CV	300
Wash 2	High conductivity	3CV	
Wash 3	Low conductivity	3CV	
Elution	Low pH	5CV	
Strip	Acidic	3CV	
Sanitization	Basic	5CV	

how the columns are washed prior to the elution. The protein A process described next was derived from a standard batch process given in Table 11.3.

11.2.3.1 Protein A Loading Zone Optimization

To maximize protein loading on the adsorbent in continuous multicolumn chromatography, columns are often configured in a series configuration in the loading zone. To further increase the specific productivity of the step, columns may also be arranged in parallel, thus splitting the feed stream and increasing the residence time in the bed. Each column or group of columns arranged in series is referred to as a stage in the adsorptive zone.

The first column in the adsorption zone (receiving fresh feed) is loaded until it is approaching its absolute capacity for protein as it exits the loading zone while each subsequent stage contains columns at reduced loadings. As such, the liquid solution contained in the interstitial space of the first column retains a considerable concentration of product, which is displaced during wash 1. To prevent this product from being lost, wash 1 is sent through either a freshly equilibrated column just entering the feeding zone, or is combined with flow-through of previous stage's effluent and sent to the final stage in the adsorption zone. BioSMB methods are typically written in the later style. There are many permutations for the loading zone configuration bounded by practical limitations, most notably pressure drop over the bank of columns, which is typically limited to 3 bar with agarose-based media due to its limited mechanical stability. Two such configurations are presented in Figure 11.11.

A design model for the protein A capture using multiple columns in the loading zone on the BioSMB has been described by Bisschops and Brower [20] and is characterized by two main parameters. The first parameter is the separation factor, S , which is the flow rate ratio of binding sites in the adsorbent (simulated) to the antibody product in the feed and represents the dimensionless loading of the adsorbent. At $S = 1$, the adsorbent is loaded exactly to its static binding capacity at equilibrium, and at values less than one, the column is loaded past its saturation point. The second parameter is the number of transfer units, NTU, which is a dimensionless unit associated with the mass transfer properties of the chromatography system. NTU can be described as the ratio between the residence time

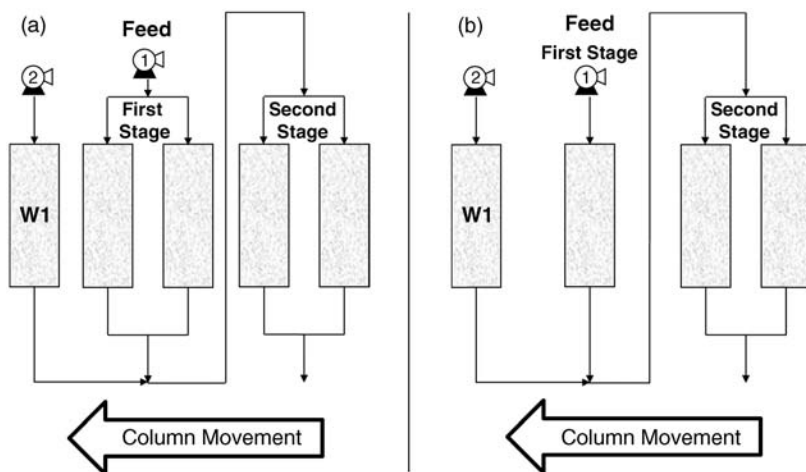


Figure 11.11 Load zone configurations for protein A monoclonal antibody capture: (a) 2 in series, 2 in parallel configuration with (4) total columns and (b) (1) column stage 1 and (2) column stage 2 with (3) total

columns. In both configurations, Wash 1 is combined with the effluent from the first stage and sent to the second stage for protein recovery.

of the liquid in the feeding zone and the characteristic time associated with the mass transfer kinetics [20]. Equations for the separation factor and the number of transfer units are given in Equations 11.6 and 11.7, respectively. The separation factor and NTU can then be used to estimate the capture efficiency, CE, or adsorption yield of the system using Equation 11.8 [23]

$$S = \frac{Q_{\text{bed}} \cdot q_m}{Q_f \cdot c_0} = \frac{V_{\text{col}} \cdot q_m}{t_s \cdot Q_f \cdot c_0} \quad (11.6)$$

$$\text{NTU} = k_{\text{oL}} a \sum \frac{N_i V_{\text{col}}}{Q_i} = k_{\text{oL}} a \left[\frac{N_1 V_{\text{col}}}{Q_f} + \frac{N_2 V_{\text{vol}}}{Q_f + Q_{\text{W1}}} \right] \quad (11.7)$$

where $Q_{\text{bed}} \equiv V_{\text{col}}/t_s$ is the simulated adsorbent phase flow rate, q_m is the static binding capacity at equilibrium, k_{oL} is the overall liquid phase mass transfer coefficient, a is the area for mass transfer, i is pass number when loading columns in series, N is the number of columns in each stage of adsorption when loading columns in series, and Q_{W1} is the flow rate of first column wash step.

$$\text{CE} = 1$$

$$- \left[\exp \left(-\text{NTU} \cdot \left(1 - \frac{1}{S} \right) \right) \cdot \left(1 - \frac{1}{S} \right) / 1 - \left(\frac{1}{S} \cdot \exp \left(-\text{NTU} \cdot \left(1 - \frac{1}{S} \right) \right) \right) \right] \quad (11.8)$$

Examination of Equations 11.6–11.8 reveals several important concepts. First, q_m and $k_{\text{oL}} a$ are the only unknown terms needing to be defined in the system. The classic chromatography concepts of adsorption isotherm analysis and frontal analysis

may be employed to estimate q_m and $k_{oL}a$, respectively. Second, the theoretical adsorption yield can be easily estimated at different operating conditions by changing the operational parameters c_0 , a function of the fermentation process, t_S , which influences S , and Q_b , which influences NTU. Q_{W1} also has a small effect on NTU.

Using this approach several theoretical operating lines have been constructed for antibody capture. The model was then validated by comparing the theoretical results with CMCC data derived from BioSMB experimental results repeated at low feed titers of 2 g l^{-1} and high feed titers of 10 g l^{-1} . The results are shown in Figure 11.12. The BioSMB data were generated using 1.6 cm diameter columns with a bed height of 2.5 cm packed with

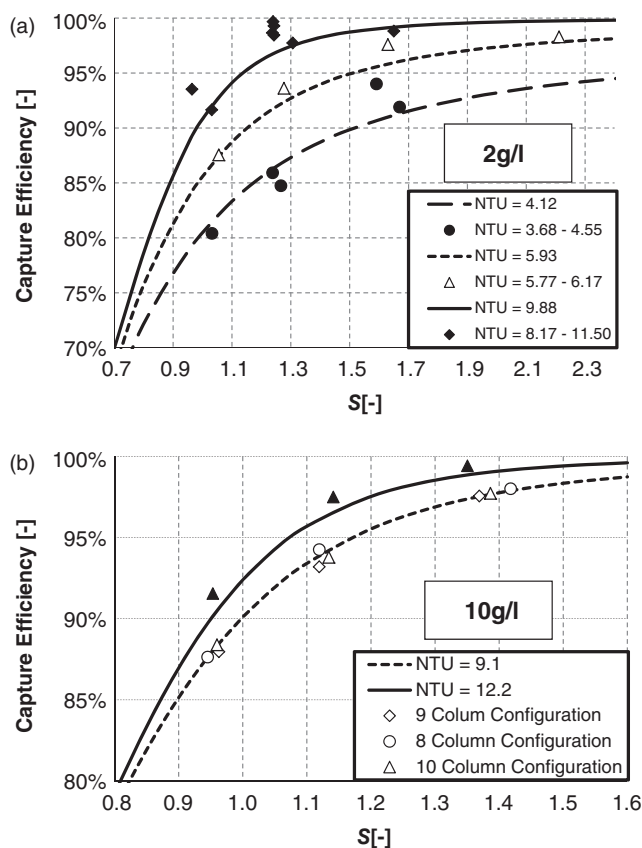


Figure 11.12 Theoretical capture efficiency curves (lines) for a monoclonal antibody on a protein A adsorbent (MabSelect SuRe™) plotted as lines with corresponding experimental data for (a) 2 g l^{-1} mAb feed stream using an 8-column ($1.6\text{ cm} \times 2.5\text{ cm}$) method and (b) 10 g l^{-1} mAb feed stream

for 8-, 9-, and 10-column methods ($1.6\text{ cm} \times 2.5\text{ cm}$). The predictive design models for BioSMB show a reasonable fit to the BioSMB experimental data within 2% as well as operational ranges where $>99\%$ capture efficiency can be achieved.

Mabselect SuReTM adsorbent. The low titer experiments were all conducted with a (4) column load zone shown in Figure 11.11a and a total of (8) columns in the method. Due the higher titer (i.e., faster switch times), and the desire to use the same column dimensions with the 10 g l⁻¹ feed, (9) and (10) column methods were conducted utilizing the (4) column load zone as well as an (8) column method using the (3) column load zone shown in Figure 11.11b. Both adding columns to the entire process, and removing a column from the loading zone, have the same effect on the method – lengthening the time available to accomplish all remaining chromatographic steps other than loading and consequently reducing the linear velocity for each of those steps.

Figure 11.11a and b demonstrate several key concepts. First, that the design equations can be used to estimate BioSMB capture performance over a wide range of *NTU* and *S* values and can be used as a starting point for experimental design. Second, that at a given flow rate, a switch time can be chosen to provide for >99% capture efficiency for either 2 g l⁻¹ or 10 g l⁻¹ feed streams. Practical considerations must always be taken into account, most importantly that as the switch time is reduced, the linear velocity of the remaining processing steps increases and the residence time decreases. The appropriate number of columns in the method must be chosen to allow for linear velocities of those remaining steps to be in the range of 300–600 cm h⁻¹. It can be seen that for a 10 g l⁻¹ feed stream, the capture efficiency is consistent regardless of the column configuration. Finally, the design equations can also be used as a basis for scale-up. Once the final titer and volume of the upstream process is known, and the desired processing time is set, the columns can be sized, and switch times chosen to give a specific *NTU* and *S*.

The degree of freedom in the BioSMB design space that allows for achieving the same capture efficiencies with multiple flow rate and switch time combinations allows for operation in different regimes. The first is characterized by heavy column cycling at separation factors between 1.25 and 1.4, and would be appropriate for clinical campaigns where it is desirable to exhaust a column's lifetime in a single campaign. The second regime operates at separation factors in the range of 1.05–1.20 where columns are loaded very close to their static binding capacity. Here cycling is less frequent and columns can support multiple campaigns, such as in the manufacturing environment. For reference, batch chromatography typically operates at separation factors between 1.5 and 1.6. It should be noted that for scale up considerations, an optimized process should hold linear velocity and residence time in each step in the process constant as is the case with traditional batch-based chromatography processes. As a result, by changing operational regimes, one would need to revalidate the process such that critical product attributes and purity requirements are maintained.

Based on the experimental data, the CMCC step was compared to the original optimized batch process on the basis of processing 2000 l of clarified cell culture fluid at different titers and the benefits are summarized in Table 11.4. The

Table 11.4 Demonstration of the options of BioSMB approaches that can provide productivity and step time improvement to conventional resin chromatography.

Protein A process	Batch	BioSMB	BioSMB	Batch	BioSMB	BioSMB	BioSMB
Titer	2 g l ⁻¹	2 g l ⁻¹	2 g l ⁻¹	10 g l ⁻¹	10 g l ⁻¹	10 g l ⁻¹	10 g l ⁻¹
Step time [h]	24	20	12	48	20	12	20
Relative total adsorbent volume [l]	1	0.26	0.43	5	0.32	0.54	0.88
Relative individual Column volume [l]	1	0.03	0.05	5	0.03	0.05	0.22
Number of columns	1	8	8	1	10	10	4
Number of cycles	4	14	9	4	54	32	17
Loading linear velocity [cm h ⁻¹]	250	440	470	250	440	470	140
Relative specific productivity [g/(l·h)]	1	3.2	3.2	0.75	14.0	14.0	4.8

CMCC process uses multiple columns 2 orders of magnitude smaller than that required for the batch process, and the CMCC process can be adapted to high titers by adding additional columns of the same dimensions as an analogous method at low titer. In addition, there are significant step time and specific productivity enhancements associated with the continuous processing scenario that are magnified at high titer. Finally, column dimension, arrangements, and step time may be manipulated to meet the desired goal of the step whether it be reduced complexity or increased cycle numbers.

11.2.3.2 Protein A Elution Zone Considerations

Protein A is bulk purification step and is not used for fine separation of closely related species. It may occur that a prepeak or tail contains the majority of impurities and should be omitted for the elution pool; however, in most instances, protein A pool cutting is designed to deliver a high recovery yield of the product. The effect of optimizing the loading zone by changing switch time in BioSMB methods results in varied flow rates and residence times for the other steps in the chromatographic process. This is of primary importance in the elution zone where mass transfer limitations exist for transporting product out of the pore structure and into the eluate stream. It may occur that the residence time is reduced to such an extent that the recovery yield is sacrificed. Figure 11.13 shows the recovery yield plotted against the residence time in the elution zone. It can be seen that at high residence times the recovery yield remains close to 100%; however, there is a critical residence time at 5 min below which yield is sacrificed. As a result, a

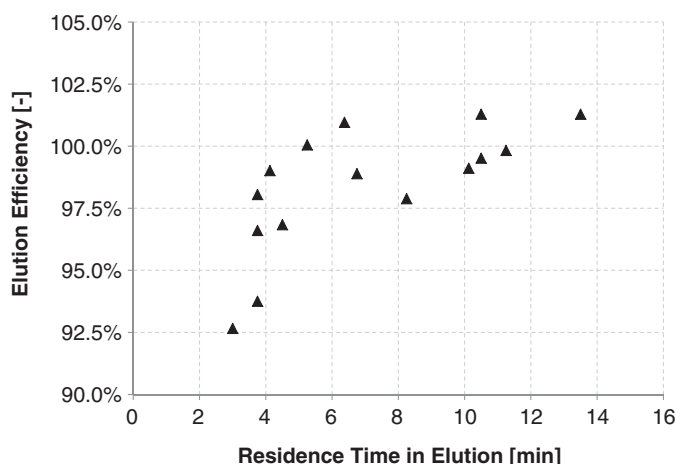


Figure 11.13 Elution efficiency of the mAb product plotted as a function of residence time in the elution zone. For this monoclonal antibody, a minimum of 5 min residence time is required to fully recover the product.

constraint is placed on the process for this molecule that the switch time must be sufficiently large to allow for 5 min residence time in the elution zone.

Processes robustness and reproducibility over a long number of cycles and multiple columns is critical to the success of a continuous multicolumn chromatography step. To ensure that the BioSMB method can accomplish a reproducible separation, a method was run across 8 days and 144 cycles. The experiment used eight 2.5 cm diameter $\times$ 2.5 cm MabSelect SuReTM columns performing the optimized method with the four-column load zone presented in Figure 11.11a. Each individual column elution was collected every 6 cycles and was assayed for yield and process residuals. The capture efficiency and elution yield were constant across the 144 cycle experiment at >99%. In addition, the clearance of process residuals also remained constant across the large number of cycles as shown in Figure 11.14. Here the data are averaged across the eight columns for each assayed cycle and there is no trend at indicating that either Chinese hamster ovary (CHO) host cell protein (CHOP) or DNA is compromised at large cycle numbers. In data not presented, it was shown that eight individual columns performed very similarly with a standard deviation of 0.06 for the average LRV DNA and CHOP clearance over the course of the experiment.

Column elutions are discrete events and as such they do not provide a constant flow rate into the next unit operation. If the peak is being collected based on UV or time triggers, portions of the eluate stream will not be brought forward into the subsequent unit operation. As such, Equation 11.9 may be used to calculate the appropriate eluate flow rate for use in the continuous process.

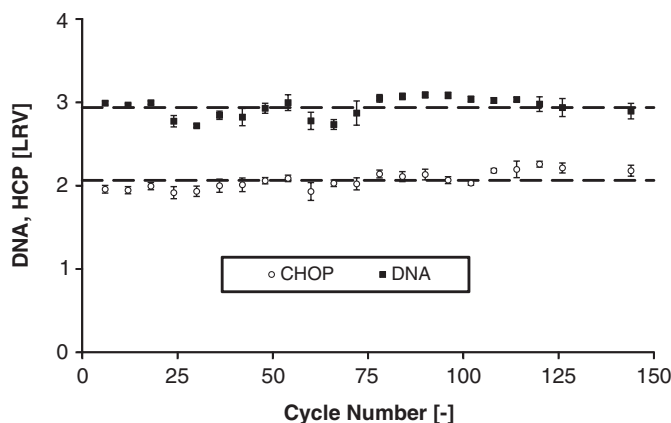


Figure 11.14 The LRV value averaged over the eight columns in the BioSMB method taken every 6 cycles for 144 cycles. Consistent residual DNA and host cell protein clearance was observed over the course of the 144 cycles of BioSMB protein A capture.

$$Q_{\text{PAP}} = \frac{V_{\text{EI}}}{t_{\text{S}}} \quad (11.9)$$

where Q_{PAP} is the protein A product flow rate into the next unit operation and V_{EI} is the total volume of protein A product brought forward into the next step.

11.2.3.3 Viral Inactivation

The US Food and Drug Administration (FDA) guidance mandates two orthogonal viral reduction processing steps in purification processes for proteins derived from mammalian cell culture [24]. Inactivation of the viral load is typically included as one of these dedicated steps and may be accomplished by several means. Effective inactivation has been shown by heat treatment and exposure to UV-C light, detergents, and solvents; however, the most common method employed in mAb purification process is a low pH shift and hold [25,26]. In typical batch processing the protein A product pool, eluted at low pH, is collected in a vessel and the pH is adjusted further down typically to $\text{pH} < 4.0$. The pool is held at that pH for defined residence time, τ_{VI} , a period of 30–60 min depending on specific viral inactivation kinetics. After the low pH hold, the inactivated protein A eluate pool is adjusted to a safe pH for protein stability.

The traditional viral inactivation process just described may be transformed into either a periodic process by repeating the batch process on a discrete number of elutions from the CMCC step, or it must be rethought of for continuous operation. One truly continuous option relies on principles developed for continuous sterilization of media into a bioreactor as shown in Figure 11.15. In such systems, nonsterile media are heated to the sterilization temperature through a heat exchanger and is maintained at that temperature

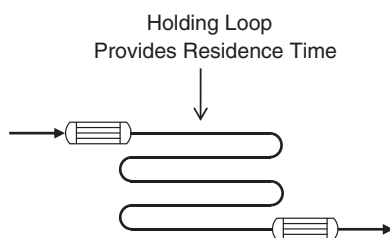


Figure 11.15 Basic schematic of a continuous sterilizer with heat exchangers to adjust the temperature of the incoming and outgoing media, and a coil to provide the desired residence time.

in a holding loop consisting of insulated lengths of pipe connected by U-bends [27]. The tube is designed to a sufficient length to give the media an appropriate residence time for sterilization. For viral inactivation of the protein A product, the same principles apply, except that the heat exchangers would be replaced with pH adjustments. The continuous pH inactivation may be accomplished using a simple silicone tubing coil to provide for a single-use solution. A starting estimation for the length of tubing required can be found using the incoming flow rate Q_{PAP} according to Equation 11.10.

$$L_{\text{VI}} = 2 \frac{(Q_{\text{PAP}} + Q_{\text{VIA}}) \cdot \tau_{\text{VI}}}{a_{\text{VI}}} \quad (11.10)$$

where L_{VI} is the length of tubing, Q_{VIA} is the flow rate of acid required to adjust the pH down for viral inactivation, and a_{VI} is the inner cross-sectional area of the desired tubing.

Care must be taken to design the tubing coil to provide the desired inactivation time on the microscopic scale and as such, Equation 11.9 is only a starting point for design calculations. The actual tube length needs to take into account any nonidealities associated with the flow in the tube, which may result in protein exiting the coil prior to τ_{VI} . As a result, coils used for viral inactivation should be well characterized by classical residence time distribution (RTD) analysis. RTD analysis refers to the amount of time each individual molecule of a trace solution spends inside the tubing coil and relies on the assumptions of constant flow rate and a Dirac input function of tracer solution to the tube. A thorough treatment of residence time distribution analysis is given by Fogler [28] as well as Froment and Bischoff [29]. An analytical solution to the tracer material balance given by Fogler [28] was used to generate the curves shown in Figure 11.16. It can be seen that as dispersion decreases in the tube, decreases, the tracer peak in the effluent becomes sharper and leaves the column later. It should be noted that tracer solution is detected in the column effluent prior to calculated residence time in the tube calculated based on flow rate and coil volume alone ($\Theta = 1$). This difference from the ideal tube residence time must be taken into account to ensure that all of the protein A product flowing through the coil is appropriately inactivated.

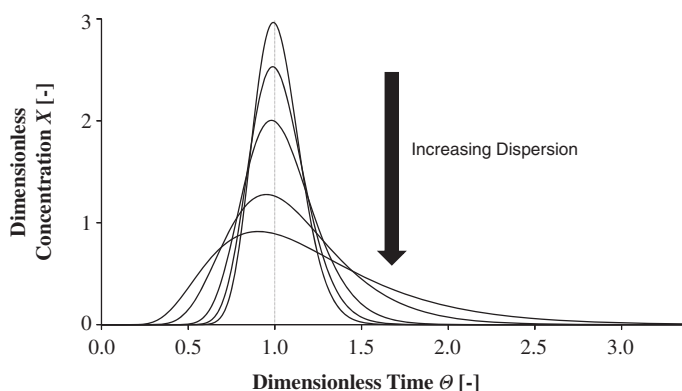


Figure 11.16 Simulated effluent curves from pulse tracer injections using the equation with increasing dispersion in the tube where θ is dimensionless time ($\theta = t/\tau_{VI} = tv/L_{VI}$), v is the superficial fluid velocity, X is the dimensionless concentration ($X = C/C_0$), and c_0 is the

concentration of the tracer solution. For RTD analysis, c_0 is defined as the mass of acetone in the pulse injection divided by the entire system volume between the injection point and the UV detector.

The target pH of the viral inactivation is set based on inactivation kinetics to provide several logs of clearance of virus – typically $>6\text{LRV}$. As such this pH set point is a critical process parameter that must be met to ensure product safety. A challenge to conducting a continuous pH adjustment is the typical pH gradient observed during the pH shift elution from the protein A columns as shown in Figure 11.17a. Here, the elution pH spans from pH6.5 to $<\text{pH}=4$. By pooling several column elutions from the CMCC step in the previral inactivation surge bag, transients in the observed pH in that vessel can be minimized, but not eliminated. Figure 11.17b shows the pH profile in the previral inactivation hold bag over a period of 20 h and represents >210 individual column elutions. Here the transients are limited to ± 0.1 pH units over time. Due to these transients, a rigorous control strategy must be put in place to ensure that only fluid with the proper pH will be sent forward into the holding loop for viral inactivation. This would include automated pH control and an automated valve system to direct flows. By using silicon tubing for the holding loop, it is possible to insert pH probes at different residence times to ensure a stable pH profile throughout the entire hold time.

The effluent of the viral inactivation coil must be properly conditioned for the next unit operation – in this instance, anion exchange chromatography. As such, the pH of the stream needs to be adjusted with a basic solution and in some instances a diluent must be added to reduce conductivity. The flow rates of the modifying solutions are accounted for in outlet flow rate of the viral inactivation step calculated as

$$Q_{\text{VIOut}} = Q_{\text{PAP}} + Q_{\text{VIA}} + Q_{\text{VIB}} + Q_{\text{VID}} \quad (11.11)$$

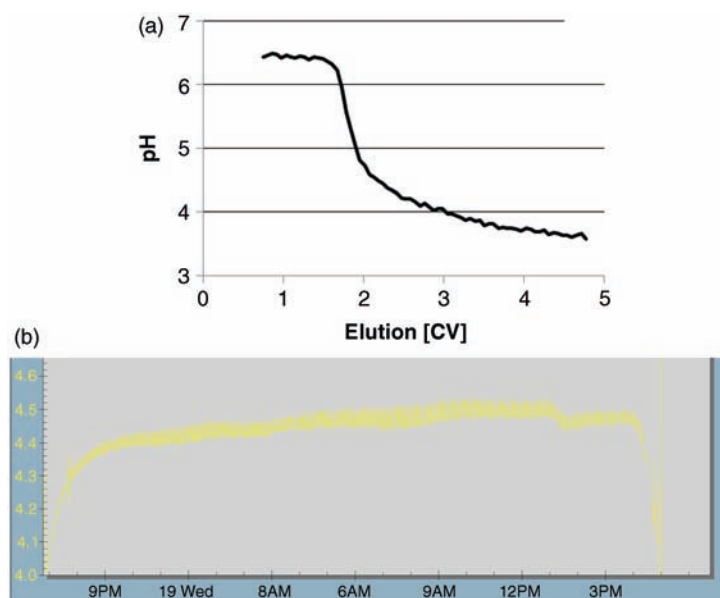


Figure 11.17 Typical pH profiles for (a) a single protein A column elution from CMCC and (b) in the previal inactivation surge bag with no pH control operated over 20 h.

where Q_{VIB} is the flow rate of base solution needed to reach the proper pH and Q_{VID} is the flow rate of diluent needed to reach the desired conductivity for the subsequent unit operation.

11.2.4

Fine Purification: Flow-Through Anion Exchange Chromatography (AEX)

Membrane chromatography is an alternative to the traditional packed-bed chromatography [24]. The advantages of using membrane adsorptive chromatography, for removal of high MW contaminants, such as DNA and viruses host cell protein (HCP), aggregates, protein A ligand, viruses, have been well established and include reduced processing time and buffer usage, higher flow rates, and ease of scale up [30–32]. Zhou showed scalability from 00.8 ml to 5 l for example. For conventional bead-based chromatography, adsorptive transport of solutes, particularly high molecular weight contaminants, such as DNA and viruses, to the binding sites relies on diffusion and is relatively slow. This mass transfer resistance leads to low efficiency with long column bed heights and slow linear flow rates on the order of $100\text{--}150\text{ cm h}^{-1}$ with dramatically oversized columns [33]. Whereas membrane adsorption relies primarily on convective mass transport with minimal pore diffusion, resulting in increased overall throughput.

Another advantage of membrane adsorption is its lower bed height to diameter ratio, which results in smaller pressure drops compared to column chromatography, which allows for flow rates on the order of 1–10 membrane volume (MV) per minute. The synthetic porous multilayered membranes are generally available in single-use formats of various configurations – discs, capsules, or cartridges – which simplifies operations (column packing eliminated) and scale-up (modular formats), and minimizes validation requirements [31]. For these modules, linear scale-up of frontal surface area, bed volume, and flow rate is possible while the dynamic binding capacity remains constant.

A variety of functionalized membranes (e.g., affinity, ion exchange, hydrophobic interaction ligands) are available on the market, but the most popular application involves the use of anion-exchange (AEX) membranes as a polishing step in the mAb purification platform. These membranes have been shown to be effective for scavenging trace amounts of impurities in flow-through mode [24]. The properties of typical AEX resin (Poros HQ50) and scale-down membrane units from Pall (Mustang Q), Millipore (ChromaSorb), and Sartorius (Sartobind Q) are shown in Table 11.5.

11.2.4.1 Effects of Sample Flow Rate on AEX Membrane Chromatography

One of the main differences between traditional and continuous processing for membrane chromatography steps is the flux through the membrane. Because the rate limiting step in continuous process is not AEX, the flow rate for the step is much lower than typical for batch-based processes. The flow rate for the continuous anion exchange (Q_{AEX}) step is equivalent to the flow rate of the effluent from the VI step such that $Q_{\text{AEX}} = Q_{\text{VIOut}}$.

The experimental results at different flow rates are summarized in Figure 11.18. At flow rates from 1 to 10 MV min⁻¹, typical for batch-based processing, there were no significant differences for both HCP level and yield. When the flow rate was reduced to 0.3 MV min⁻¹ for the continuous processing scenario, the HCP level was reduced compared to the higher flow rates (>75 ppm) while still maintaining a high recovery yield (>95%). It is suggested that operation in continuous process mode at reduced flux values could significantly enhance the impurity removal. It should be noted however that the increased residence time in the membrane adsorber may decrease the yield of the step with some molecules as a result of the timescale of binding kinetics. To date, this phenomenon has not been observed experimentally by the author.

11.2.4.2 Effect of Sample Loading Amount on AEX Membrane Chromatography

Sample loading is another significant factor for the process optimization of AEX membrane chromatography. In order to determine the sample loading quantity, a dynamic binding capacity study was carried out at the optimal pH, conductivity, and flow rate conditions. Figure 11.19 shows the breakthrough curves for both HCP and DNA concentrations at low flow (0.3 MV min⁻¹). The typical loading for membrane adsorbers depends on the quality of the feed material, but

Table 11.5 Summary of properties for anion exchange bead based resin and membrane adsorber units.

Name	Resin	Membrane		
	POROS HQ50	Pall Mustang Q Coin	Millipore ChromaSorb	Sartobind SingleSep Q
Sorbent/ device				
Unit size	2.0 ml	0.35 ml	0.08 ml	1.0 ml
Pore size	0.36 µm	0.8 µm	0.65 µm	>3 µm
Chemistry	Quaternized polyethyleneimine	Quaternary ammonium	Polysallyamine	Quaternary amine
Support matrix	Cross-linked poly (styrenedivinylbenzene)	Polyethersulfone (PES)	Ultrahigh molecular weight polyethylene (UPE) membrane	Stabilized reinforced cellulose

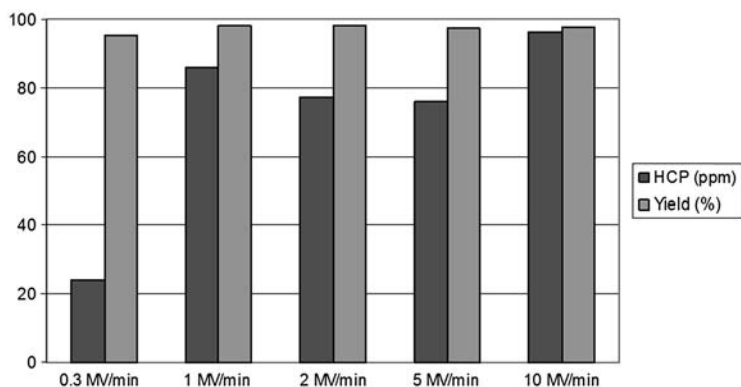


Figure 11.18 Comparison of Pall Mustang Q membrane adsorber performance for host cell protein concentration (ppm) and yield (%) in the flow through pool as a function of flow rate.

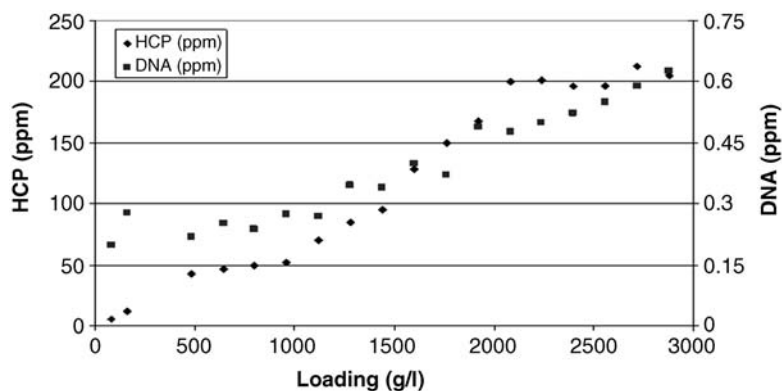


Figure 11.19 HCP (ppm) and DNA (ppm) concentration as a function of Pall Mustang Q membrane adsorber loading at optimal pH and conductivity values.

is typically in the range of $1000\sim3000\text{ g l}^{-1}$, which is approximately 10 times greater than traditional AEX column loadings for the same separation ($\sim100\text{ g l}^{-1}$).

11.2.4.3 Scaling-Up Membrane Chromatography for Continuous Processing

In traditional AEX column, the loading linear velocity and column bed height are conserved upon scale-up, but for membrane chromatography, flux is held constant because the number of membrane layers (analogous to bed height) remains the same regardless of the total membrane volume in the module. In a case study, three scales of membrane processes were performed with Pall Mustang Q adsorbers. The experimental conditions and analytical results for all three scales are summarized in Table 11.6. A $3\times$ buffer reduction was demonstrated in this case study using a much smaller device (260 ml membrane vs. 2.5 l resin) as

Table 11.6 Summary of results of Mustang Q membrane at different scales.

Scale	Small (MV = 0.35 ml)	Medium (MV = 60 ml)	Large (MV = 260 ml)
Flow rate	0.3 MV min ⁻¹	0.1 MV min ⁻¹	0.1 MV min ⁻¹
Feed HCP	343.5 ppm	92.7 ppm	818.6 ppm
Feed DNA	2.2 ppm	0.7 ppm	7.9 ppm
Yield	95.1%	95.6%	>95%
HCP	24.0 ppm LRV: 1.16	9.8 ppm LRV: 0.98	57.8 ppm LRV: 1.15
DNA	1.3 ppm LRV: 0.28	0.2 ppm LRV: 0.54	1.6 ppm LRV: 0.70
Monomer	n/a	n/a	98.85%

compared to the AEX column. The scale-up study also shows more than 95% step yield with HCP <60 ppm and DNA <2 ppm in the flow-through pool at all scales using Mustang Q adsorbers with membrane volumes of 0.35, 60, and 260 ml. (Note: the designed flow rate was 0.1 MV, but it was not feasible to test in the small scale Mustang Q coin (0.35 ml) due to flow rate limitations). In the continuous process, the flow rate for the AEX step ($Q_{\text{AEX}} \equiv Q_{\text{AEXIn}} \equiv Q_{\text{AEXOut}}$) is equal to Q_{VIOut} .

11.2.5

Fine Purification: Continuous Multicolumn Chromatography – BioSMB Cation Exchange Chromatography

Continuous multicolumn chromatography can be used for other bind and elute modalities, such as cation exchange chromatography. Cation exchange is traditionally used in antibody processes to separate closely related species to the product, such as aggregates and charged variants. Product may be eluted from cation exchange columns by using either pH or conductivity gradients or steps. A pH adjustment of the cation exchange feed stream to acidic conditions is necessary to protonate proteins in solution, which promotes binding to the negatively charged adsorbent. The CMCC step development performed on the BioSMB was similar to the approach presented in Sections 11.2.3.1–11.2.3.2 for protein A chromatography.

11.2.5.1 Cation Exchange Loading Zone Optimization

The loading zone for cation exchange chromatography was designed in either of the two configurations presented in Figure 11.20. In Figure 11.20a, feed is uninterrupted as it was presented for protein A chromatography. In Figure 11.20b, the feed is periodically suspended during the time Wash 1 is sent to the second stage for product recovery. Both load zones were tested for capture efficiency using (8) or (6) columns (1.2 cm diameter × 3.4 cm, Poros HS50) for the two different load zones accordingly. A design model was created for CEX following a similar approach to that defined for protein A capture using Equations 11.6–11.8. The

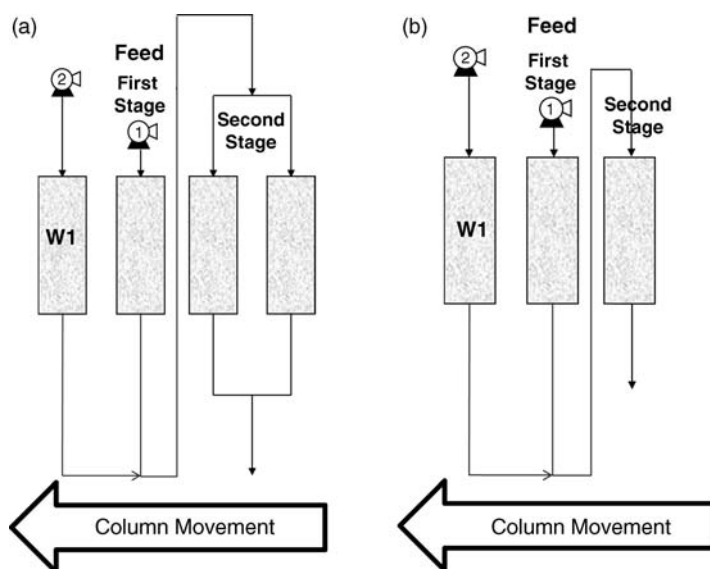


Figure 11.20 BioSMB Load zone configurations for cation exchange monoclonal antibody capture: (a) (1) column stage 1 and (2) column stage 2 with (3) total columns, and (b) (1) column stage 1 and (1) column in stage 2 with (2) total columns. In configuration (a),

Wash 1 is combined with the effluent from the first stage and sent to the second stage for protein recovery. In configuration (b), feed is suspended as Wash 1 is sent to the second stage for recovery.

capture efficiency model predictions are shown in Figure 11.21. As with protein A chromatography, the models provide a good fit to the experimentally derived data (within 1%) regardless of load zone configuration. Due to the reduced number of columns and the possibility to match cycle times with the protein A step, the configuration presented in Figure 11.20.b was selected as the preferred loading zone configuration.

11.2.5.2 Cation Exchange Elution Zone Considerations

After binding conditions were finalized, the elution zone of the cation exchange process was optimized to achieve a similar separation compared to the batch-based processes. Since cation exchange chromatography is providing fine separations of product-related proteins (aggregates, fragments, variants, etc.) it requires tight control with high reproducibility. However, the requirement to use smaller chromatography columns with reduced bed heights in continuous multicolumn chromatography to gain efficiency in the loading zone may lead to detrimental effects in the elution zone due to reduced total plate counts and residence times. As a result, the resolution of critical components must be verified upon transition to a CMCC process. The cation exchange step is primarily responsible for separating mAb aggregate from monomer; as such, aggregate clearance was utilized to demonstrate comparability of CMCC to the reference batch process.

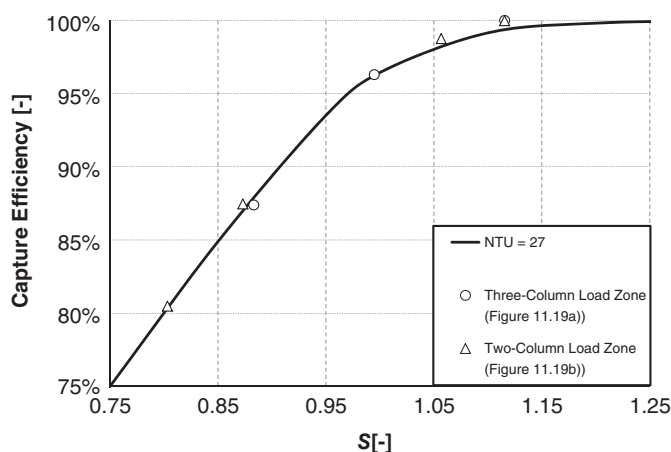


Figure 11.21 Theoretical capture efficiency curves (lines) for a monoclonal antibody on a cation exchange adsorbent (Poros HS50) plotted as lines with corresponding experimental data for (a) 10 g l⁻¹ mAb feed stream using a

six-column (1.2 cm × 3.4 cm) method. The predictive design models for BioSMB show a good fit to the BioSMB experimental data within 1% as well as operational ranges where >99% capture efficiency can be achieved.

The aggregate clearance profiles presented in Figure 11.22 were generated on standard batch columns (0.5 cm diameter × 20 cm), reduced bed height batch columns (1.2 cm diameter × 6.8 cm), and with the continuous chromatography configuration using (6) 1.2 cm diameter × 3.4 cm columns all packed with Poros

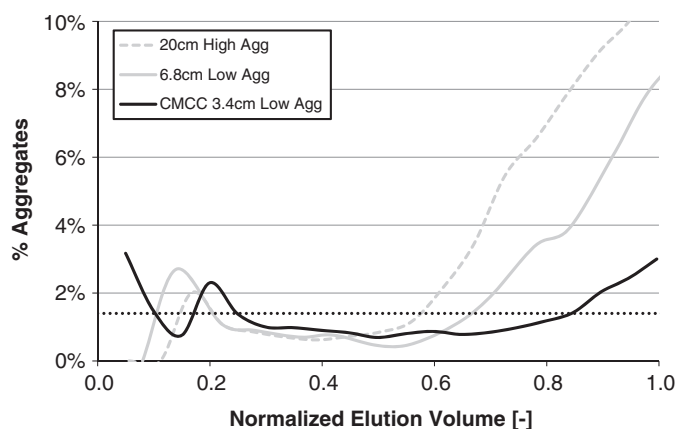


Figure 11.22 Aggregate elution profiles from cation exchange chromatography using Poros HS50 media. The continuous chromatography results are compared with batch chromatography results using two different column

configurations; 0.5 cm diameter × 20 cm and 1.2 cm diameter × 6.8 cm. The CMCC separation is comparable to the batch-based separation. The dotted horizontal line represents the aggregate percentage in the feed.

HS50 adsorbent. It can be seen that aggregate clearance profiles of all three scenarios are similar with a small aggregate peak eluting near the beginning of the profile as well as an increasing aggregate percentage as elution proceeds. The aggregates are held to a lower final percentage in the CMCC CEX pool compared to the batch alternative. The position of the leading aggregate peak in the profiles is shifted for the different curves as a result of varying system voidages and fraction sizes for each experiment. In addition, the falling aggregate percentage seen at the onset of the CMCC elution is carried over from the preceding cycle in the eluate line and is not included in the product pool. The separation achieved with the CMCC system was determined to be comparable or better than the reference batch-based process. The similar aggregate trends observed at different bed heights in this work is in part due to the nature of the Poros media, which allows for convective flow throughout the bead and relies less on diffusive mass transport. Similar performance may not be observed on more traditional agarose-based media.

In multicolumn chromatography, peak collection can be accomplished by traditional means, that is, UV, pH, conductivity peak characteristics or time/volume based. In the case of the cation exchange chromatography step described earlier, time-based collected was chosen as the preferred method. Using those cut points, the CMCC process was scaled up to 2.5×5 cm columns for 16 cycles. The UV traces from those cycles (overlaid) are shown in Figure 11.23. Over the course of the 16 cycles, the standard deviation of peak area for each column was $<0.5\%$ and the standard deviation among all six columns was 1% demonstrating the robustness of the process.

11.2.6

Formulation: Continuous Ultrafiltration

In recent years, single-pass tangential-flow filtration (SPTFF) has been suggested as an alternative traditional multipass tangential-flow filtration (MPTFF) for protein concentration. In SPTFF, stacks of ultrafiltration membranes are arranged in such a manner as to create stepwise changes in the flowpath, which results in a consistent channel velocity throughout a single membrane cassette [34,35]. Constant retentate flow rate in the membrane channel is considered a key scaling rule in conventional multipass tangential-flow filtration because it determines the effectiveness of the sweeping action of the fluid flow at the membrane surface. This sweeping action is responsible for minimizing the deleterious effects of the concentration polarization layer, a build-up of solute compounds including ions, macromolecules, colloids, and the protein product at the membrane surface, on flux [34].

Traditional tangential-flow filtration is a low conversion process and as a result, the retentate must be passed through the membrane cassette many times before the final concentration factor (CF) can be achieved. As a result, a recirculation tank and pump are considered mandatory in these systems. Alternatively, single-pass tangential-flow filtration is a high conversion process requiring only

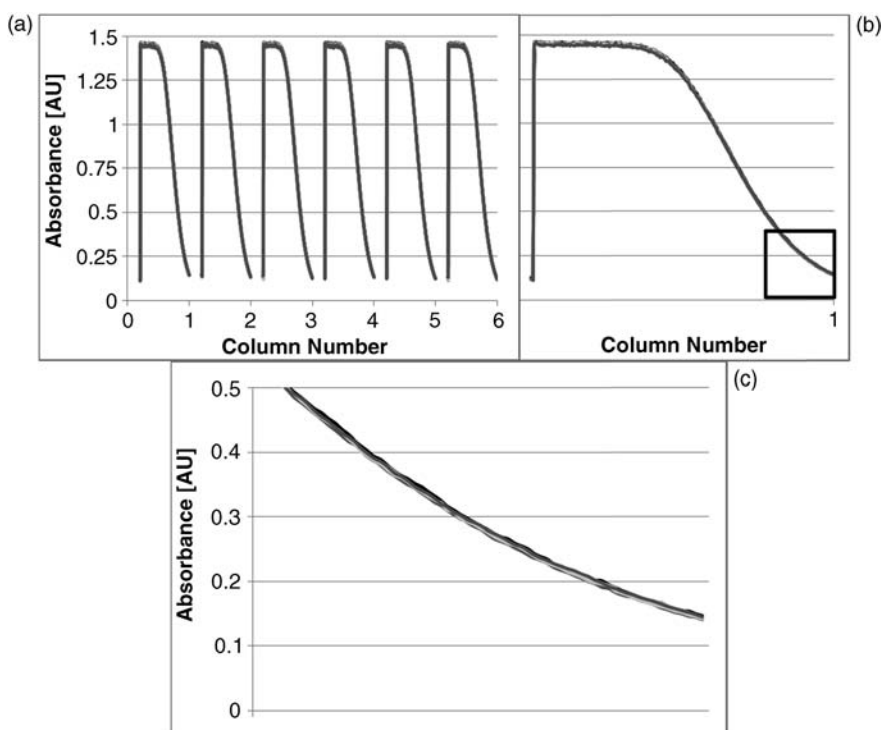


Figure 11.23 UV (280 nm) elution profiles from 16 cation exchange cycles overlaid for (a) all six columns used in the CMCC method, (b) a close-up of just column 1 elutions, and (c) a zoomed-in region of column 1 elutions at the tail edge of the peak. All elution peak areas were within 1% standard deviation.

one pass through the membrane cassette. Process flow diagrams for MPTFF and SPTFF are presented in Figure 11.24a and b, respectively. Initial applications for SPTFF have been focused on three main areas: (i) high titer concentrations where the reduced system hold-up and efficient product recovery are advantageous, (ii) systems with shear sensitive proteins or foamy streams where the lack

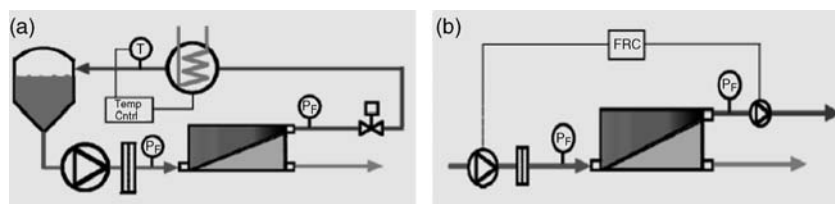


Figure 11.24 Block flow diagram for (a) traditional MPTFF showing the large recirculation tank and pump required for the concentration and (b) SPTFF showing the flow ratio control logic (FRC) required to meet the desired concentration in a single pass [36].

of recirculation can minimize the deleterious effects on product quality or flux decay, and (iii) as a flow-through concentration device used to de-bottleneck processes due to tankage limitations. Similar to the third application, the nature of the SPTFF mode of action is also amenable to continuous operation where the incoming stream is relatively dilute, and the outgoing retentate is at the target concentration.

The final ultrafiltration step may also be responsible for formulating the product into drug substance by the addition of a diafiltration stage of the filtration either in the middle or at the end of the concentration phase of the process. In MPTFF, diafiltration is a simple operation of either: (i) feeding and bleeding the diafiltration buffer into the recirculation tank at a rate matching the permeate flow rate, or (ii) by a series of 5–8 successive whole volume additions of the buffer to the recirculation tank followed by a concentration back to the original volume. In addition, due to the low efficiency per pass, the final concentration can be finely tuned by extending the filtration time. To date single-pass tangential-flow filtration suffers from the inability to accomplish the diafiltration in the same single-pass through the membrane cassette and also relies on the RFC control loop to finely tune in the final protein concentration. By placing a conventional MPTFF, solely responsible for diafiltration and concentration following the SPTFF step, both of these issues are eliminated. The continuous process can run with some variation in the concentration factor (within bounds) and final formulation can be done offline of the continuous equipment train using a relatively small filtration skid with low hold-up volumes.

The concentration factors that can be achieved in single-pass tangential-flow filtration are a factor of four principal parameters. First is the length of the channel and as such SPTFF membranes are sold in cassettes of varying configurations (4-in-series, 7-in-series, 9-in-series) to extend the flow path while maintaining the linear velocity in the channel. Second is the flow rate; longer residence times leads to higher concentration factors. The third parameter is the inlet and outlet pressures, which are a function of flow rate and manually applied retentate backpressure. The final parameter is the feed concentration, which has implications on the concentration polarization layer formation. To understand the interdependence of the first three key operating parameters listed, an experiment was performed on a partially purified monoclonal antibody solution with an incoming concentration of 9.3 g l^{-1} . The solution was passed through the SPTFF apparatus at different inlet pressures (i.e., transmembrane pressures as the retentate pressure = 0 psi) and flow rates (i.e., residence times) with two different membrane configurations. The results of this experiment are presented in Figure 11.25a and b for a 4-in-series (0.065 m^2) and 7-in-series (0.12 m^2) cassette configurations. Because the inlet pressure, outlet pressure, and flow rate are intrinsically linked, the operating space becomes complex in that a desired concentration factor can be achieved by multiple combinations of flow rate and pressures. For this experiment, a 5 l of feed solution was placed in a bottle along with the retentate and permeate

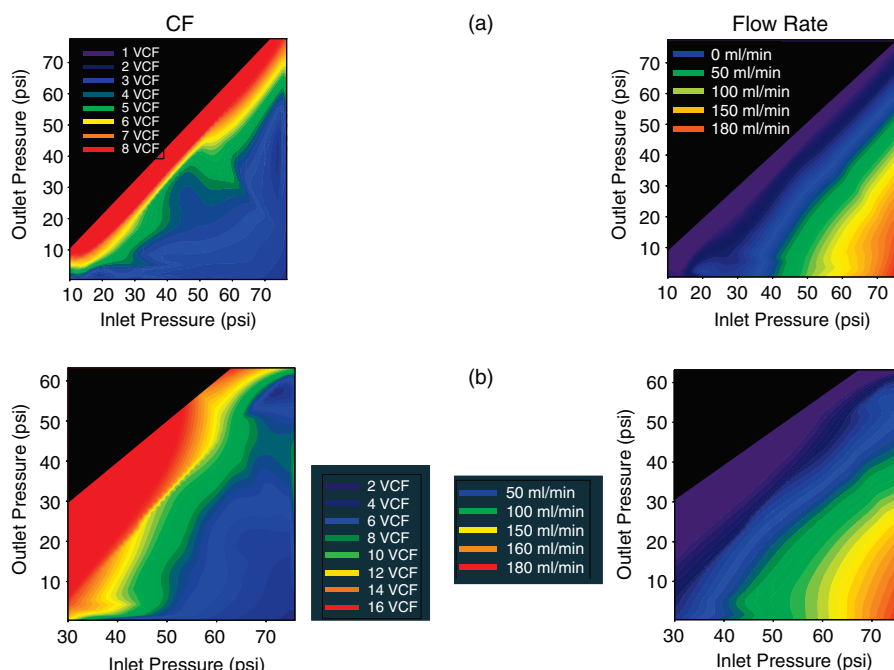


Figure 11.25 Single-pass tangential-flow filtration data (a) 4-in-series cassette with 0.65 m² membrane area and (b) 7-in-series cassette with 0.12 m² membrane area. The left panes are surface plots of the concentration

factor as a result of the inlet and outlet pressures. The right panes are surface plots of the flow rate as a result of the inlet and outlet pressures. Data generated by Taylor Chartier at Merck Research Labs, Rahway, NJ.

resulting in total recycle of the process streams. After conditions were set, the SPTFF system was run for 15 min to establish equilibrium before recording the concentration factor and pressure information. In the data presented in Figure 11.25, the concentration factor for each cassette can be selected in the left-hand pane thus setting inlet and outlet pressures. Those pressures can be translated to the right-hand pane to determine the operating flow rate (or the reverse).

The incoming flow rate to the SPTFF is set by the preceding anion exchange chromatography step such that $Q_{\text{SPTFFIn}} \equiv Q_{\text{AEX}}$. The SPTFF is then set to concentrate to a specific volumetric concentration factor (VCF) such that

$$Q_{\text{SPTFFIn}} = Q_{\text{SPTFFR}} + Q_{\text{SPTFFP}} \quad (11.12)$$

Where Q_{SPTFFR} is the retentate flow rate and Q_{SPTFFP} is the permeate flow rate, and $VCF \equiv Q_{\text{SPTFFIn}}/Q_{\text{SPTFFR}}$.

The SPTFF was not originally designed for use on continuous processes where incoming flow rates are at the laboratory scale (<200 l bioreactors) where typical incoming flow rates are often <50 ml min⁻¹, hence the relatively large cited

membrane areas. As a result, the permeate flow is often restricted in the continuous processing scenarios to prevent overconcentration of the stream in the cassette and the formation of a gel layer at the membrane surface. This issue would be eliminated if appropriate scale-down membranes were available on the market. At larger scales, current membrane offerings become more appropriate for continuous processing purposes.

11.3

Pilot-Scale Demonstration of the Integrated Continuous Process

The demonstration of the continuous process at the pilot scale was conducted with 100 l of cell culture fluid produced in a single-use bioreactor with a product titer of $\sim 2 \text{ g l}^{-1}$. The crude cell broth was processed through the following unit operations continuously: kSep Single-Use Centrifugation $\rightarrow$ Depth Filtration $\rightarrow$ Protein A Capture Chromatography (CMCC BioSMB) $\rightarrow$ Low pH Viral Inactivation $\rightarrow$ Anion Exchange Chromatography (Membrane Adsorber) $\rightarrow$ Cation Exchange Chromatography $\rightarrow$ Continuous Ultrafiltration. To achieve this, each of the individually optimized unit operations described in Section 11.2 was scaled-up individually and connected through the use of surge bags at the appropriate flow rates. This setup is shown in Figure 11.26a while the time each step in the continuous DSP was in active operation is shown in Figure 11.26b. The unit operations were started up sequentially starting with the centrifugation step and ending with the continuous ultrafiltration. The unit operation then ran for a

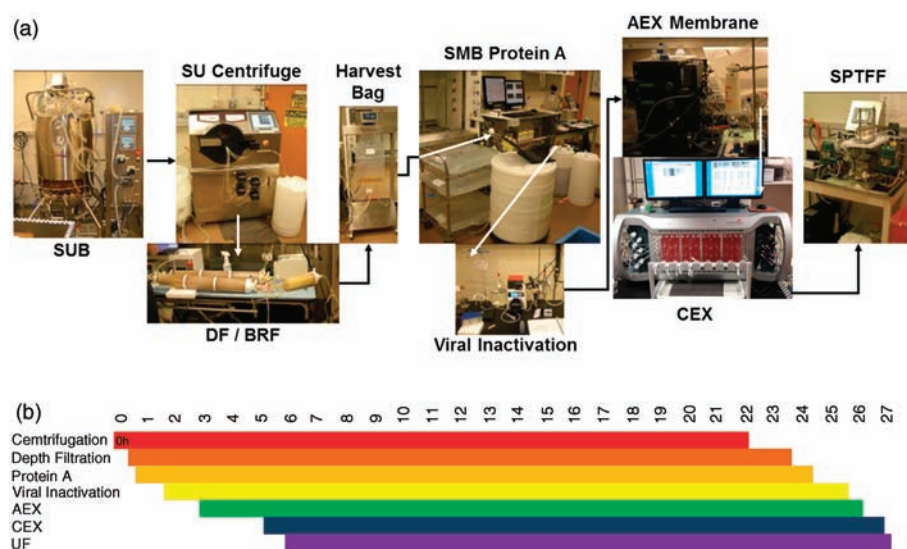


Figure 11.26 (a) Pilot setup of the first-generation 100 l continuous purification demonstration and (b) unit operation usage schedule.

Table 11.7 Summary of performance for the proof of concept pilot scale continuous processing of CHO expressed mAb with the single use enabled platform.

	Average yield [%]	DNA* [ppm]	HCP [ppm]	Res. ProA [ppm]	% Monomer
Centrifugation	97.3	N/S	N/S	N/S	N/S
DF/BRF	98.6	30 515	383 300	N/S	N/S
Protein A SMB	98.1	N/S	N/S	N/S	N/S
Viral inactivation	100	2	1063	2.1	89.8%
Anion exchange membrane	98.8	<LOQ	82	1.5	99.0%
Cation exchange chromatography	84.2	<LOQ	605	<LOQ	99.2%
SPTFF	99.5	0.001	35	<LOQ	99.0%
Overall	77.9	0.001	8.7	<LOQ	99.0%

period of ~18 h of continuous, uninterrupted operation before the unit operations were turned off sequentially as feed became depleted. The timescale of the process was set to harvest the batch through protein A chromatography over 24 h thus setting the timescale for the rest of the DSP. Future scale-up work will reduce the protein A harvest time such that the entire purification could be completed within a 24 h time period; requiring slightly different column dimensions in the protein A and cation exchange chromatography steps.

Each of the surge bags servicing the unit operations was sampled three times throughout the continuous downstream process demonstration. Titer analysis as well as HCP and DNA assays were performed on each of the samples. Table 11.7 summarizes the performance of each of the unit operations averaged over the course of the demonstration.

Surge vessels provide the analogous point in batch processing where intermediates are assayed for critical product attributes and purification progression. At a minimum, such analytic capabilities are currently required in the bag to measure any critical process parameters associated with the succeeding unit operation, such as pH and/or conductivity. However, a longer-term process analytic technology (PAT) objective for a continuous process will be to use online and rapid at-line analytical techniques to sample each bag at periodic intervals for process residuals, titer, and other quality-related methods Figure 11.27. Feedback into a multivariate data analysis (MVDA) system can then be incorporated to let operators know when a process starts going out of control so that corrective action may be taken. Furthermore, a detailed Failure Mode Analysis must be conducted to engineer in sophisticated controls that mitigate against loss of a batch from an individual out of control event. Out-of-spec material as measured by the PAT approach must be prevented from moving downstream in the continuous process and eliminate the possibility for contaminating all of the material processed prior to loss of control.

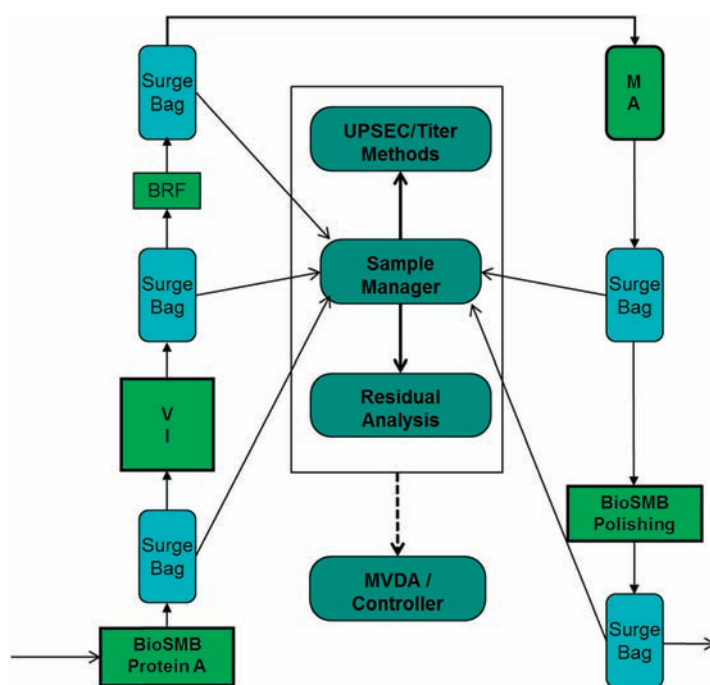


Figure 11.27 Future state of process monitoring and control. Each surge bag is sampled frequently for quality and process residuals. MVDA can then be applied to the continuous downstream process to alert operators if the process starts going out of control.

11.4

Summary

This work has provided the proof of concept for a single-use enabled continuous downstream processing for the next-generation CHO mAb manufacturing. The economic analysis shows that this approach can provide a significant economic favorable advantage over conventional stainless steel approaches. Combining this 2000l single-use closed processing modular “ball room” type modular facility of the future will enable flexible processing solutions that can handle the fluctuations of capacity demand from blockbuster needs to smaller personal medicine type demands. The continuous processing provides rapid processing options that will accelerate not only commercial manufacturing but clinical manufacturing as well where speed to clinical studies is key. Continuous processing provides significant capital savings from the elimination of hold vessels as outlined in Figure 11.28. The move toward commercial implementation still requires significant development investment. This needs the development of automation approaches that integrate process models with PAT. In addition the robustness of single-use technology needs to be improved especially the reduction of leak issues by the

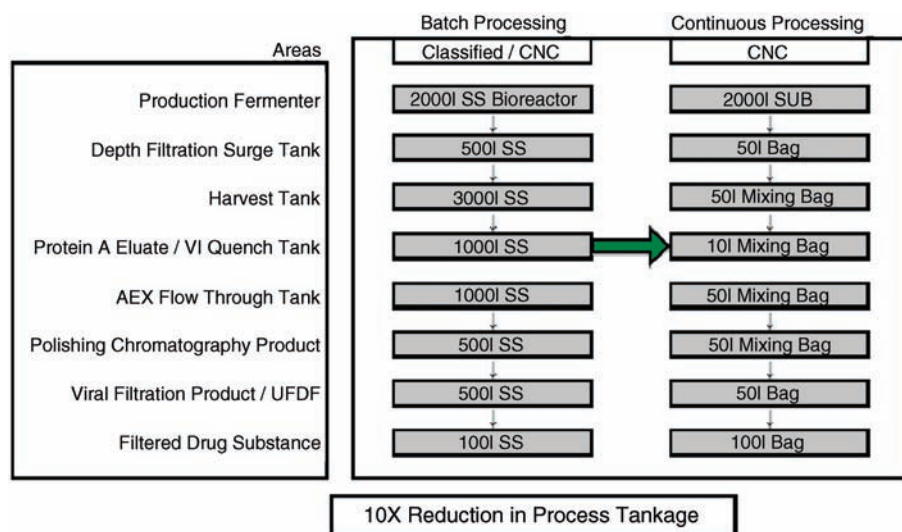


Figure 11.28 The impact of continuous processing to reducing tank volumes and providing a potential fivefold reduction in capital avoidance.

introduction of molded tubing/bag manifolds and supply chain transparency. These issues including improvements to commercial implementation, such as defining guidance for regulatory and GMP approaches for leachables and particulates, are being tackled by the end-user consortium of the BPOG disposables working group (biophorum.com). In addition, appropriate continuous processing scale-down tools need to be enhanced and integrated to enable rapid process development of continuous processing. It is predicted that this approach will provide a significant impact to the next-generation processing in the next 5 to 10 years.

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